Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement 51 - Virtual Technical Conference Platform

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GitHub Repository:

https://github.com/bhaveshv24/SE-Labs-PES1UG24CS115.git

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Requirements Table

Problem Statement 51 - Virtual Technical Conference Platform

Actors: Conference Author, Track Chair

Functional Requirements

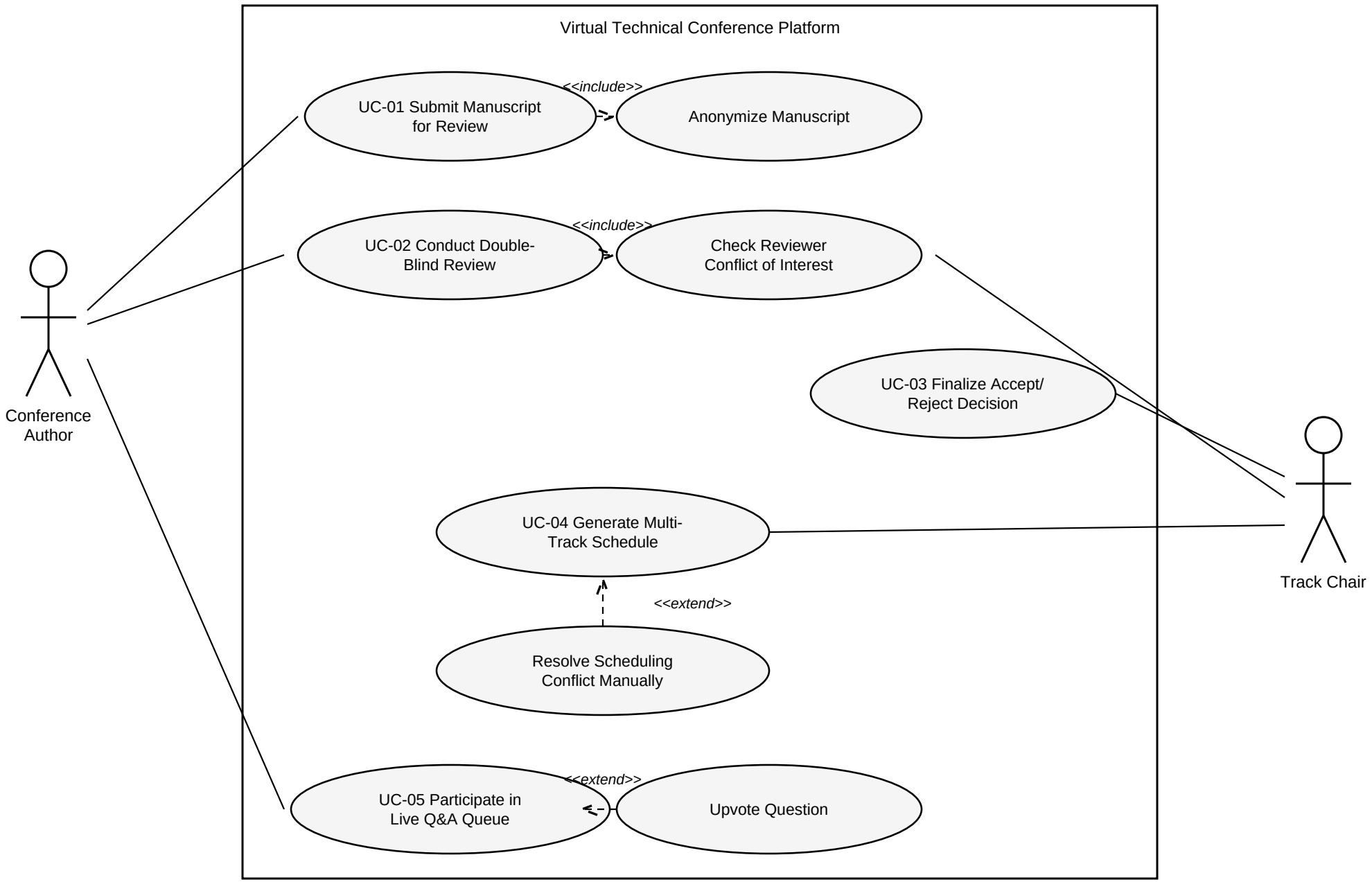
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall anonymize submitted manuscripts (remove author name, affiliation, and file metadata) before assigning them to reviewers for double-blind review.	High	Pass: reviewer view of paper shows no author name/affiliation. Fail: author identity visible to reviewer.	Needed for double-blind review process.
FR-002	Functional	The system shall assign each paper to reviewers based on topic area while excluding reviewers who have a declared conflict of interest with the author.	High	Pass: no paper is assigned to a conflicted reviewer and every paper gets the minimum number of reviewers. Fail: a conflicted reviewer is assigned or paper has too few reviewers.	Avoids biased reviews.
FR-003	Functional	The system shall generate a conflict-free multi-track schedule that respects room capacity, track timings and speaker availability.	High	Pass: no room/speaker double-booking and no session over capacity in generated schedule. Fail: any conflict present.	Given directly in problem statement.
FR-004	Functional	The system shall let attendees submit questions during a live session and upvote existing questions, with the queue reordering by upvote count.	Medium	Pass: new question appears in queue within 3 sec and queue reorders after upvote. Fail: not reflected in time.	Needed for live Q&A feature mentioned in problem statement.
FR-005	Functional	The system shall notify the Track Chair once all reviews for a paper in their track are complete.	Medium	Pass: Track Chair gets notification within 5 min of last review being submitted. Fail: no notification sent.	Lets Track Chair finalize decision without manually checking.

Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional	The multi-track schedule generator shall resolve room and speaker	High	Pass: benchmark shows generation completes in	Given in problem statement as

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
		constraints in under 10 seconds under simulated peak load.		under 10 sec at peak load. Fail: exceeds 10 sec.	sample NFR.
NFR-002	Nonfunctional	The live Q&A queue shall stay available and correctly ordered for at least 500 concurrent attendees with 99.5% uptime during session hours.	Medium	Pass: load test with 500 users shows correct ordering and no dropped submissions; uptime logs meet 99.5%. Fail: ordering breaks or uptime below target.	Q&A is audience-facing during live sessions so reliability matters.

Use Case Diagram - Virtual Technical Conference Platform



Use-Case Flow Specification

Virtual Technical Conference Platform - Problem Statement 51

Use Case: Conduct Double-Blind Review

Use Case Name	Conduct Double-Blind Review
Actors	Track Chair (primary), Conference Author (secondary - via Submit Manuscript for Review)
Related Requirements	FR-001, FR-002, FR-005
Brief Description	Enables a Track Chair to have a submitted, anonymized manuscript evaluated by a conflict-free pool of qualified reviewers, culminating in the reviewers' recommendations being recorded for the accept/reject decision.
Preconditions	• The manuscript has been successfully submitted through Submit Manuscript for Review and has passed the Anonymize Manuscript step (author identity and metadata stripped). • The Track Chair is authenticated and has permission to manage reviews for the track the paper is assigned to. • A pool of registered reviewers with declared topic-area expertise exists for the paper's track.
Postconditions	• Success: The paper has the configured minimum number of completed, non-conflicted reviews, and the Track Chair is notified that the paper is ready for a final decision. • Failure: The paper remains in a pending-review state and no notification is sent to the Track Chair.
Main Success Scenario	1. Track Chair opens the list of anonymized papers pending review for their track. 2. System shows each paper's topic area(s) and current reviewer-assignment status. 3. Track Chair selects a paper and requests reviewer assignment. 4. System identifies candidate reviewers whose expertise matches the paper's topic area(s). 5. System checks each candidate for a conflict of interest with the author(s) (include: Check Reviewer Conflict of Interest). 6. System excludes conflicted reviewers and assigns the paper to the configured minimum number of remaining qualified reviewers. 7. Each assigned reviewer is notified and accesses the anonymized manuscript via their review queue. 8. Each reviewer submits a structured review (comments, score, recommendation). 9. System records each review against the paper and tracks completion status. 10. Once all reviews are in, system marks the paper "Ready for Decision" and notifies the Track Chair (FR-005). 11. Use case ends successfully.
Alternate Flow	**A1 - Insufficient Qualified Reviewers Available** Triggered at step 6, if fewer than the minimum required non-conflicted reviewers can be found for the paper's topic area(s). A1.1. System flags the paper "Reviewer Shortage" instead of completing assignment. A1.2. System notifies the Track Chair that manual intervention is required. A1.3. Track Chair broadens the topic-area match or invites an external reviewer to the pool. A1.4. Track Chair re-triggers reviewer assignment (resumes at step 4). A1.5. If reviewers still can't be found, the Track Chair may proceed with a reduced count, logged as a policy exception.
Special Requirements	• Reviewer identity must not be exposed to the Conference Author at any point in this flow (double-blind, not single-blind). • Conflict-of-interest checks must run before manuscript access is granted to any candidate reviewer, per FR-002.